

SDP 302

Problem Set: Multicollinearity & PCA

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Instructions & Information

The Longley dataset is a small macroeconomic dataset widely used in statistics, econometrics, and data science. Introduced by Longley [1967](#) to assess computational precision, it captures the macroeconomic landscape of post-WWII US economic growth and demographic expansion across just 16 rows and 7 variables. Almost all variables exhibit near-perfect co-movement, making the dataset notorious for high inter-variable correlation. Consequently, it serves both as a classic stress test for statistical modeling and a prime example for severe multicollinearity and numerical instability. Despite its small size for modern standards, the Longley dataset remains popular due to its simplicity and accessibility. It comes pre-packaged in many statistical libraries and features in intermediate and advanced textbooks, including Greene [2012](#) or Heiberger and Holland [2015](#), amongst others.

Based on macroeconomic demand theory, Longley used the following multiple linear regression equation to model post-WWII US employment:

$$\begin{aligned} \text{TOTEMP}_t = & \beta_0 + \beta_1 \text{GNPDEFL}_t + \beta_2 \text{GNP}_t + \beta_3 \text{UNEMP}_t + \beta_4 \text{ARMED}_t \\ & + \beta_5 \text{POP}_t + \beta_6 \text{YEAR}_t + \epsilon_t \end{aligned}$$

When standard Ordinary Least Squares (OLS) regression is applied directly to this dataset, strong correlations among the independent variables lead to inflated variance, unstable coefficient estimates, and numerical instability. Principal Component Analysis (PCA) is an unsupervised technique used to address these issues. It simplifies high-dimensional data by transforming correlated variables into a smaller set of uncorrelated variables called the principal components, which retain as much of the original variance and information as possible. Constructed as linear combinations of mean-centred variables using eigenvectors, principal components are orthogonal, i.e. statistically independent, by design. As a result, PCA also helps to uncover underlying structural relationships that might otherwise remain hidden. For more information on PCA, the book by James et al. [2013](#) provides a thorough treatment of both the theoretical framework and practical use cases. In Python, PCA is part of the statsmodel library.

References

- Greene, William H. (2012). *Econometric Analysis*. 7th. Boston: Prentice Hall.
- Heiberger, Richard M. and Burt Holland (2015). *Statistical Analysis and Data Display*. 2nd. Springer.
- James, Gareth et al. (2013). *An Introduction to Statistical Learning*. Vol. 103. Springer Texts in Statistics. New York, NY: Springer New York.
- Longley, James W. (1967). “An Appraisal of Least Squares Programs for the Electronic Computer from the Point of View of the User”. In: *Journal of the American Statistical Association* 62.310, pp. 319–341.

[EXR] Exercise 1 (Exploring the BLUE Properties of OLS)

The Longley dataset is part of Python's statsmodel library and can easily be loaded calling the method `statsmodel.datasets.longley.load_pandas()`. Import the data into a DataFrame and answer the following questions:

- a) Display the data in a well-formatted table (e.g., using pandas or tabulate). What structural patterns or trends do you observe across the variables?
- b) Based on macroeconomic theory, what are the expected signs (+ or -) for each coefficient in Longley's regression equation?
- c) Fit the OLS regression model. Do the estimated coefficients match your expected signs? Do you notice any surprising results (e.g., unexpected p-values or signs)? What might explain them?
- d) Compute and display the correlation matrix for the six independent variables. Also for each variable calculate the variance inflation factor. What conclusions can you draw about variable redundancy?
- e) Normalise each independent variable using 1947 as the base year. Generate matrix scatter plots of all pairwise combinations, and plot the trends of all six variables from 1947 to 1962 in a single chart. What patterns emerge?
- f) Apply PCA to reduce dimensionality and eliminate correlation. Can you offer a meaningful interpretation for each of the primary principal components?